DSA06 DATA HANDLING AND VISUALIZATION | Activity: Map Storytelling Challenge | Slot D

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Time: 20 minutes Total: 20 marks Tool: R (ggplot2 + maps). Your map story is different from your neighbour's; in the last 3 minutes, swap stories with a partner and say what you would change.

Story Brief MS-D02: Illiteracy (% of population, 1970) [20 marks]

Brief: You are building a map story about Illiteracy (% of population, 1970) across the United States for a classroom poster. Your comparison variable is Frost days (days per year below freezing). Your focus region is the South region. Your map uses 6 classes and the "plasma" colour scale. Statistics use all 50 states; the map shows the 48 contiguous states (Alaska and Hawaii are not in map_data).

Set-up: run this code first (do not edit it).

```
library(ggplot2); library(maps)
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost", "area") d$id
<- tolower(d$state)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]
V <- d$illit # the variable you map
W <- d$frost # the comparison variable
```

Tasks:

1. The base map. Draw a choropleth of illit: ggplot(mapdat, aes(long, lat, group = group, fill = illit)) + geom_polygon(colour = "white") + coord_map("albers", lat0 = 29.5, lat1 = 45.5) + scale_fill_viridis_c(option = "plasma") with a clear title and legend [1]. Report the highest state and its value [1], the lowest value and the state(s) that have it [1], and the median of all 50 states [1]. [4]
2. The regional story. Use tapply(V, d$zone, mean). Give the mean for each of the four regions [1.5] and name the highest and lowest region and the gap between them [1]. Then compute cor(V, W), describe its strength (weak below 0.3, moderate 0.3 to 0.7, strong above 0.7) and direction in one sentence, noting that it is not proof of cause [1.5]. [4]
3. Classes change the story. Make two versions of the map with 6 classes: equal interval, cut(V, 6), and quantile, cut(V, quantile(V, seq(0, 1, length.out = 7)), include.lowest = TRUE). For each, report the lower limit of the top class and the number of states in it [2.5]. Which method makes the highest values look rarer, and which would you choose for your audience? [1.5] [4]
4. Counts versus rates. Create the number of illiterate people: count <- d$pop * 1000 * d$illit / 100. Name the top 3 states by count and the top 3 states by illit [2]; say how many states appear in both lists [1]. In one sentence, explain why a map of counts would mislead and what to map instead [1]. [4]
5. Tell the story for the South region. Report the mean of illit in South and its ratio to the mean of all 50 states [1]; the highest state in South with its value [0.5]; and how many South states are above the national median [0.5]. Then write a headline of at most 12 words [1], and two evidence sentences using your numbers plus one recommendation for a classroom poster [1]. [4]

Good map stories: *one clear message, honest classes, rates not raw counts, a readable legend, and a source note.*

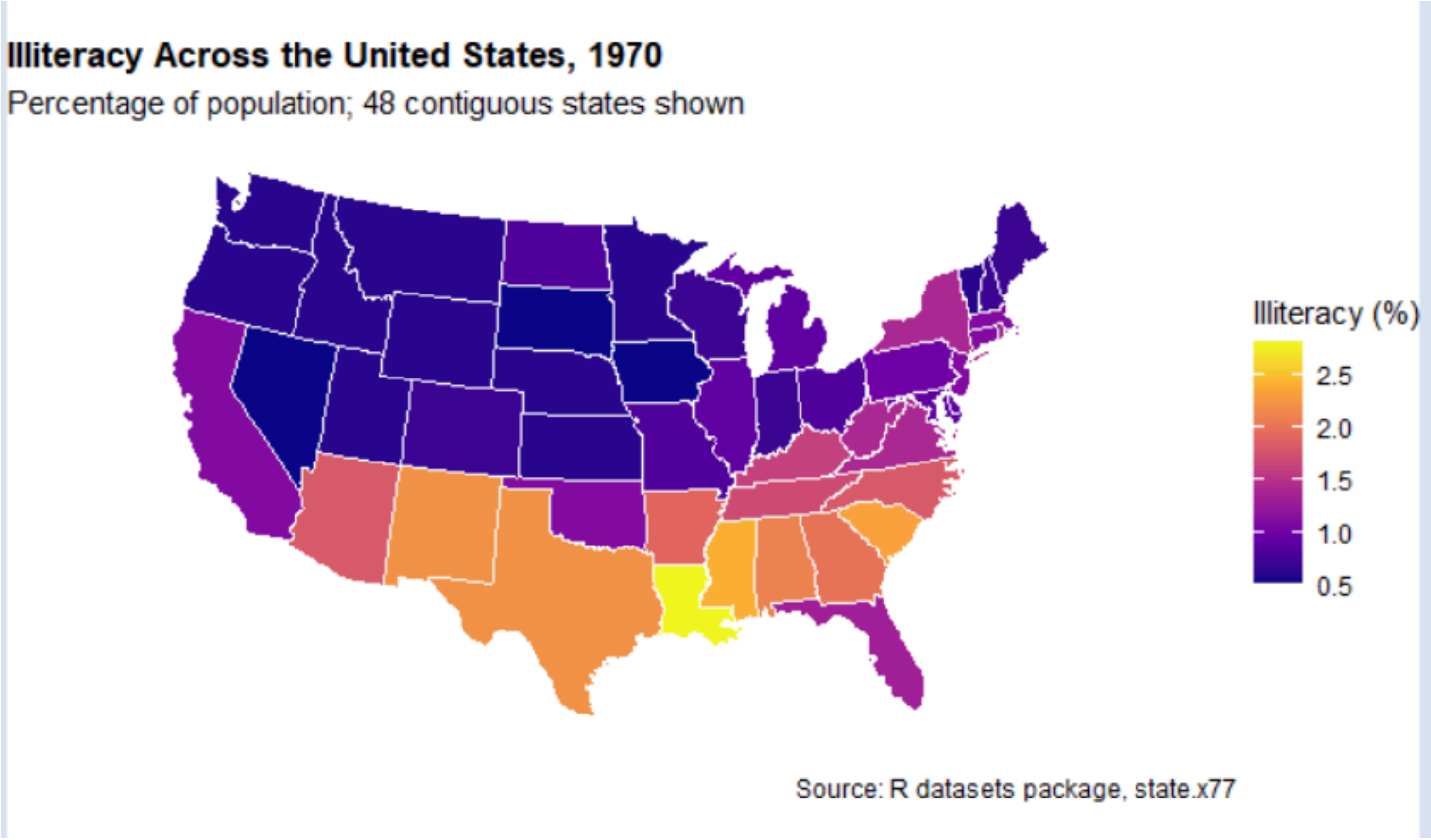
Code of Conduct: I certify that this work is my own. Signature: _____

SOLUTION:

1. The base map [4 marks]

```
library(ggplot2)
library(maps)
# Set up the supplied dataset
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit",
              "life", "murder", "hsgrad", "frost", "area")
did <- tolower(dstate)
mapdat <- merge(map_data("state"), d,
                by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdatgroup, mapdatorder), ]
```

```
V <- dillitW <- dfrost
# Base choropleth map
p1 <- ggplot(mapdat,
             aes(long, lat, group = group, fill = illit)) +
  geom_polygon(colour = "white", linewidth = 0.2) +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_c(option = "plasma",
                       name = "Illiteracy (%)") +
  labs(title = "Illiteracy Across the United States, 1970",
       subtitle = "Percentage of population; 48 contiguous states shown",
       caption = "Source: R datasets package, state.x77",
       x = NULL, y = NULL) +
  theme_void() +
  theme(plot.title = element_text(face = "bold"),
        legend.position = "right")
print(p1)
# Required statistics: all 50 states
cat("Highest illiteracy state:",
    d$state[which.max(V)], "\n")
cat("Highest illiteracy value:", max(V), "\n")
cat("Lowest illiteracy value:", min(V), "\n")
cat("State(s) with lowest value:",
    paste(d$state[V == min(V)], collapse = ", "), "\n")
cat("National median:", median(V), "\n")
```



Interpretation: The darkest or brightest areas on the plasma scale represent states with higher illiteracy percentages. The map displays the contiguous states, while the summary statistics include all 50 states.

2. The regional story [4 marks]

```
# Mean illiteracy by region
regional_mean <- tapply(V, d$zone, mean)
cat("\nMean illiteracy by region:\n")
print(round(regional_mean, 2))
highest_region <- names(which.max(regional_mean))
lowest_region <- names(which.min(regional_mean))
regional_gap <- max(regional_mean) - min(regional_mean)
cat("Highest region:", highest_region, "\n")
cat("Lowest region:", lowest_region, "\n")
cat("Gap between regional means:",
    round(regional_gap, 2), "percentage points\n")
# Correlation between illiteracy and frost days
r <- cor(V, W)
cat("Correlation between illiteracy and frost:", round(r, 3), "\n")
strength <- if (abs(r) < 0.3) {
  "weak"
} else if (abs(r) <= 0.7) {
  "moderate"
} else {
  "strong"
}
direction <- if (r > 0) "positive" else "negative"
cat("Strength:", strength, "\n")
cat("Direction:", direction, "\n")
```

```
>
> cat("Strength:", strength, "\n")
Strength: moderate
> cat("Direction:", direction, "\n")
Direction: negative
>
```

Answer format: The region with the highest mean illiteracy is the one with the largest value in the output, while the region with the lowest mean has the smallest value. The gap is the difference between these means.

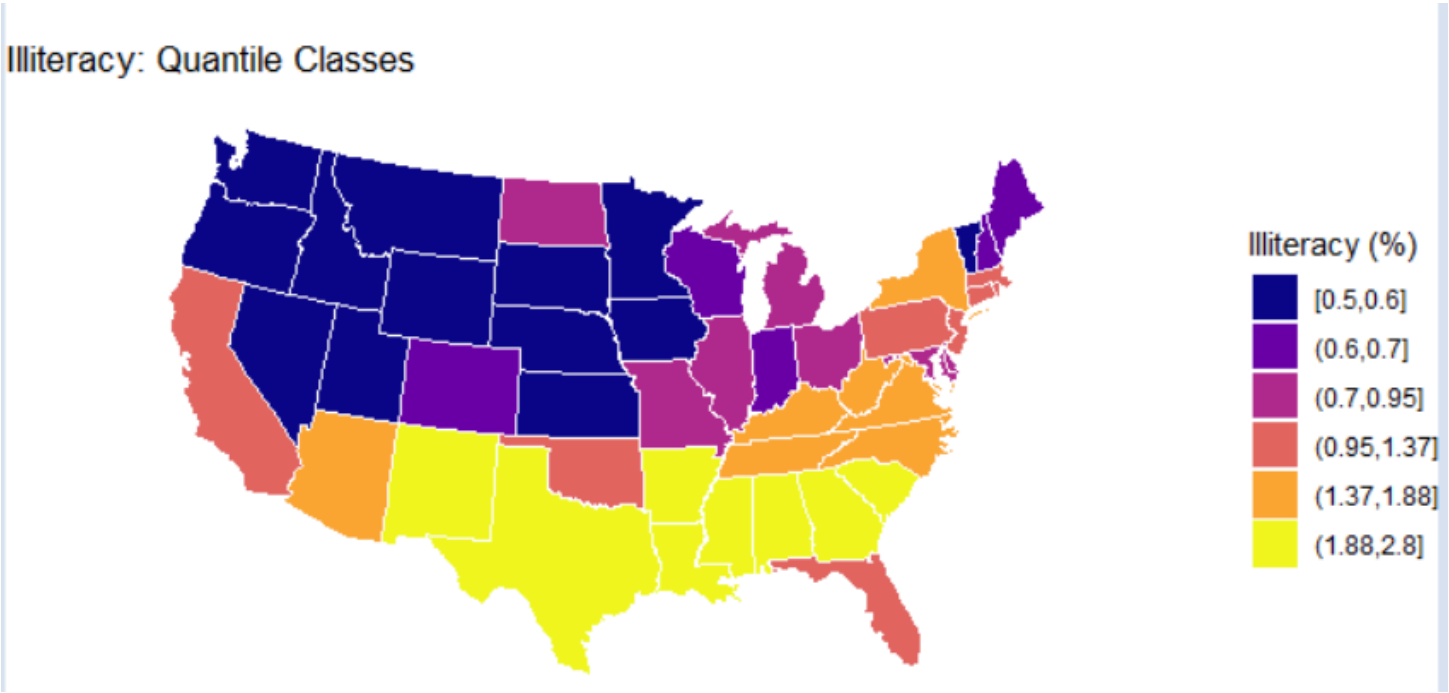
Correlation sentence: Illiteracy and frost days have a [weak/moderate/strong] [positive/negative] correlation (r = value), but this association does not prove that frost causes illiteracy.

3. Classes change the story [4 marks]

```
# Six equal-interval classes
equal_class <- cut(V, 6)
# Six quantile classes
quantile_breaks <- quantile(
```

```
  V, seq(0, 1, length.out = 7)
)
quantile_class <- cut(
  V, quantile_breaks, include.lowest = TRUE
)
# Calculate top-class lower limits and counts
top_class_summary <- function(x, classes, method) {
  top_level <- tail(levels(classes), 1)
  lower_limit <- as.numeric(
    sub("^[[(]([^,]+),.*", "\\1", top_level)
  )
  count <- sum(classes == top_level, na.rm = TRUE)
  data.frame(
    Method = method,
    Top_class_lower_limit = lower_limit,
    States_in_top_class = count
  )
}
class_results <- rbind(
  top_class_summary(V, equal_class, "Equal interval"),
  top_class_summary(V, quantile_class, "Quantile")
)
cat("\nTop-class comparison:\n")
print(class_results)
# Create maps for the two classification methods
mapdat$equal_class <- cut(mapdat$illit, 6)
mapdat$quantile_class <- cut(mapdat$illit,
  quantile_breaks,
  include.lowest = TRUE
)
p_equal <- ggplot(mapdat,
                  aes(long, lat, group = group,
                      fill = equal_class)) +
  geom_polygon(colour = "white", linewidth = 0.2) +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_d(option = "plasma",
                       name = "Illiteracy (%)") +
  labs(title = "Illiteracy: Equal-Interval Classes") +
  theme_void()
p_quantile <- ggplot(mapdat,
                     aes(long, lat, group = group,
                         fill = quantile_class)) +
  geom_polygon(colour = "white", linewidth = 0.2) +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_d(option = "plasma",
                       name = "Illiteracy (%)") +
  labs(title = "Illiteracy: Quantile Classes") +
```

```
  theme_void()
print(p_equal)
print(p_quantile)
```



Interpretation: Equal-interval classification uses the same numerical width for every class, whereas quantile classification attempts to place a similar number of states in each class. Quantile classes can make the highest values appear less rare because the top class is designed to contain a share of the states.

Recommendation: For a classroom audience, I would choose equal-interval classes when the goal is to show the actual size of differences in illiteracy percentages. I would choose quantile classes when the goal is to compare state rankings and relative position.

4. Counts versus rates [4 marks]

```
# Estimate the number of illiterate people
count <- dpop * 1000 * dillit / 100
top_count_indices <- order(count, decreasing = TRUE)[1:3]
top_rate_indices <- order(V, decreasing = TRUE)[1:3]
top_count_states <- dstate[top_count_indices]top_rate_states <- dstate[top_rate_indices]
cat("\nTop 3 states by estimated count:\n")
print(data.frame(
  State = top_count_states,
  Estimated_count = round(count[top_count_indices])
))
cat("\nTop 3 states by illiteracy percentage:\n")
print(data.frame(
  State = top_rate_states,
  Illiteracy_percentage = V[top_rate_indices]
))
overlap <- intersect(top_count_states, top_rate_states)
cat("Number of states in both top-three lists:",
```

```
   length(overlap), "\n")
cat("Overlapping states:",
   if (length(overlap) > 0) paste(overlap, collapse = ", ")
   else "None", "\n")
```

```
Top 3 states by illiteracy percentage:
> print(data.frame(
+   State = top_rate_states,
+   Illiteracy_percentage = V[top_rate_indices]
+ ))
            State Illiteracy_percentage
1       Louisiana                   2.8
2     Mississippi                   2.4
3 South Carolina                    2.3
>
> overlap <- intersect(top_count_states, top_rate_states)
> cat("Number of states in both top-three lists:",
+     length(overlap), "\n")
Number of states in both top-three lists: 0
> cat("Overlapping states:",
+     if (length(overlap) > 0) paste(overlap, collapse = ", ")
+     else "None", "\n")
Overlapping states: None
```

Answer: A count-based map can mislead because states with larger populations may have more illiterate people even if their illiteracy percentage is lower. To compare the prevalence of illiteracy fairly between states, map illiteracy percentage rather than raw counts.

5. Tell the story for the South region [4 marks]

```
# South-region statistics
south <- d[dzone == "South",]south_mean <- mean(southillit)
national_mean <- mean(V)
national_median <- median(V)
south_ratio <- south_mean / national_mean
south_highest_index <- which.max(southillit)south_highest_state <- southstate[south_highest_index]
south_highest_value <- southillit[south_highest_index]south_above_median <- sum(southillit > national_median)
cat("\nSouth-region story:\n")
cat("Mean illiteracy in South:",
   round(south_mean, 2), "%\n")
cat("National mean illiteracy:",
   round(national_mean, 2), "%\n")
cat("South mean / national mean:",
   round(south_ratio, 2), "\n")
cat("Highest South state:", south_highest_state, "\n")
cat("Its illiteracy value:", south_highest_value, "%\n")
```

```
cat("South states above national median:",
    south_above_median, "\n")
cat("National median:", round(national_median, 2), "%\n")
```

```
South-region story:
> cat("Mean illiteracy in South:",
+     round(south_mean, 2), "%\n")
Mean illiteracy in South: 1.74 %
> cat("National mean illiteracy:",
+     round(national_mean, 2), "%\n")
National mean illiteracy: 1.17 %
> cat("South mean / national mean:",
+     round(south_ratio, 2), "\n")
South mean / national mean: 1.49
> cat("Highest South state:", south_highest_state, "\n")
Highest South state: Louisiana
> cat("Its illiteracy value:", south_highest_value, "%\n")
Its illiteracy value: 2.8 %
> cat("South states above national median:",
+     south_above_median, "\n")
South states above national median: 14
> cat("National median:", round(national_median, 2), "%\n")
National median: 0.95 %
>
>
```

Suggested map-story write-up

Headline:

Illiteracy Across the South: Regional Differences in 1970

Evidence sentence 1: The South's mean illiteracy rate was [South mean]%, which was [ratio] times the national mean.

Evidence sentence 2: [Highest South state] recorded the highest South-region illiteracy rate at [value]%, and [number] South states were above the national median.

Recommendation: Use the plasma choropleth with a clearly labelled percentage legend, six-class comparison maps, and a source note identifying state.x77. Explain that the map shows percentages, not the number of people affected.